Yingqi Zhang

Balchelor of Science Mathematics

Residence: Room 2004, Building 4, Xueqing Yuan, Haidian District, Beijing, China
Phone: +1 (301)3106498, +86 17512549674 **Email:** yzhan106@terpmail.umd.edu

Summary

Bachelor student in Mathematics and currently focused on data analysis, synthetic data, causal analysis and data selection. Experienced in mathematical deviations, financial time series data analysis, remote HPC environment management, causal analysis and data selection with influence functions. Currently enrollled in a research internship with professor Steve (Xue) Liu at MBZUAI.

Education

University of Maryland College Park **College Park, Maryland, USA**
Bachelor of Science Mathematics, Certificate in Leadership Studies *Sep 2021 - April 2025*

- Relevant Coursework: Advanced Mathematics, Complex Variables Analysis, Probability, Applied Linear Algebra, Numerical Analysis, Partial Differential Equations

Skills

Technical Focus: Mathematical Modeling, Data Analysis, Causal Machine Learning
Methods: Time Series Data Analysis, Mathematic Deviations, Synthetic Data Generation, Numerical Analysis
Programming Languages: Python, C++, Java
Tools and OS: Github/Git, Linux/Macos

Publications

Nonlinear Aggregation Data Attribution (In Progress for ICLR 2027)

- We propose NADA, a nonlinear aggregation data selection framework that evaluate data on their contribution to ordinary data selection methods.

Amortized Regime-Aware World Models for Support-Preserving Adaptation in Nonstationary POMDPs (In Progress for ICLR 2027)

- We proposes AmoRAWM, a regime-aware world model for nonstationary POMDPs that adapts online by estimating the current hidden regime from a short recent context and maintaining a recurrent belief-like latent state.

Projects

Latent Severity Scores for Causal Adjustment in Early Right-Heart Catheterization (Published at HIS 2026)

- We estimated the effect of early right-heart catheterization on 30-day survival with modeled illness severity using physiologic and laboratory proxy variables.

AI-HITS: A Closed-Loop AI-Enabled Hospital Triage and Resource-Orchestration System (HIS 2026)

- We propose a closed-loop architecture that connects multi-modal sensing, safety- constrained AI decisions, coordinated execution, and outcome feedback to effectively speeds up the average waiting time for patients in hospital.

Research and Professional Experience

Mohanmod Ben Zayed university of Artificial Intelligence **Abu Dhabi, UAE**
Research Internship, Professor Steve Liu, Professor of Computer Science and Machine Learning *May 2026 – Aug 2026*

- Research topic: Shapley Value for AI Agents
- Learned to derive and apply mathematical concept from Shapley Values to Influence Function, preparing to produce a publication in related fields.
- Develop skills such as fine-tuning, Cluster management and complex mathematics implementations.

The State University of New York at Stony Brook **Stony Brook, NY**
Research Assistant, Professor Xianfeng Gu, Dept. of Computer Science and Applied Mathematics *Jun 2024 – Aug 2024*

- Learned to read academic mathematics papers and conducted research on Euclidean surface approximations
- Implemented paper algorithms in C++ for convex caps and Euclidean surfaces, converting 3D scanned models into 2D surface outputs to reduce storage requirements

Morgan Stanley **Beijing, China**
Internship in FOF department *Jun 2023 – Aug 2023*

- Created a Python algorithm to visualize financial element data and conduct comparative analysis against specified targets
- Authored eight financial reports of approximately five pages each on emerging industries, including video games, Chinese chips, and social media platforms, to support investment evaluation